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The role of artificial intelligence in enhancing firm investment efficiency

Nan Lin [a], Ao Li [b,*], Yi Geng [b], Junchen Yao [c], Chengyi Liu [d,*]

[a] School of Finance and Accounting, Fuzhou University of International Studies and Trade, China
[b] School of Finance, Zhongnan University of Economics and Law, China
[c] School of Materials and Microelectronics, Wuhan University of Technology, China
[d] School of Finance and Economics, Jimei University, China

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ABSTRACT

This paper investigates the effect of artificial intelligence (AI) on firm investment efficiency. The results show that AI can significantly enhance the investment efficiency of enterprises. This effect is more pronounced in high knowledge-intensity firms, high R&D-intensity firms, and firms with sound internal controls. Mechanism analysis reveals that AI contributes to improved investment efficiency by enhancing innovation capabilities and strengthening the financial health of firms. Furthermore, we find that AI adoption may lead to a rise in firm value. This study adds new insights to how AI influences firms' financial and strategic decision-making, providing a valuable reference for policymakers. It offers guidance for the government in formulating policies that promote intelligent transformation and encourage enterprises to leverage AI for innovation and improved investment efficiency.

1. Introduction

Artificial intelligence (AI) is one of the most significant achievements of 21st-century science and technology and a key driver of progress in the information age. Today, AI plays a leading role in industrial transformation, technological innovation, and economic modernization, profoundly impacting the economy, society, and everyday life. By the end of 2024, China had more than 4500 AI companies, with the core AI industry valued at nearly 600 billion RMB. Moreover, China's authorities have released multiple strategic guidelines, the "Smart Manufacturing Development Plan (2016–2020)", emphasizing the importance of intelligent production technologies in driving the manufacturing industry's shift toward high-quality growth.

In recent years, investment activity among Chinese manufacturing firms has steadily increased; however, enhancing investment efficiency continues to pose major challenges. Notable regional imbalances persist, and a significant number of enterprises operate below optimal efficiency levels. As key drivers of the real economy, these firms must improve their investment efficiency—not only to ensure their own long-term viability but also to support the broader objective of fostering high-quality national economic growth. Prior research has mainly focused on issues such as weak internal controls, information asymmetry, limited access to financing, and agency conflicts as determinants of investment inefficiency (Denis & Sibilkov, 2010; Liu & Yu, 2023; McLean, Zhang, & Zhao, 2012; Stein, 2003; Zheng & Xiao, 2020). While artificial intelligence (AI) has attracted increasing attention in corporate strategy and performance discussions, its specific impact on investment efficiency remains insufficiently examined. This study seeks to explore how the adoption of AI technologies influences the efficiency of corporate investment decisions. Using panel data from Chinese A-share listed companies between 2008 and 2023, we also analyze the mechanisms through which AI may alter firms' investment behavior.[1]

This paper argues that artificial intelligence enhances corporate investment efficiency through two primary mechanisms. First, AI boosts investment efficiency by strengthening firms' innovation capabilities. One of AI's most significant advantages is its ability to extract deep

* Corresponding author.
E-mail addresses: linnannv@fzfu.edu.cn (N. Lin), ali@zuel.edu.cn (A. Li), lcychengyi@gmail.com (C. Liu).

[1] The Chinese context offers a unique and highly relevant setting for examining the impact of AI adoption on firm-level investment efficiency due to its distinct institutional, economic, and policy characteristics. Unlike many Western economies, China has pursued a top-down, state-led approach to technological development, with national initiatives such as "Made in China 2025" and the "Smart Manufacturing Development Plan (2016–2020)" serving as strong policy signals that actively shape corporate behavior. Additionally, China's corporate landscape features a large number of firms, significant regional disparities, and higher levels of information asymmetry—particularly among SMEs—resulting in more severe investment inefficiencies. In this environment, AI's potential to reduce frictions and enhance decision-making is likely to manifest differently than in more mature and transparent markets, making China an ideal context for this study.

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